STATISTICAL PROCESS CONTROL ANALYSIS OF AUTOMOTIVE BRAKE ROTOR MANUFACTURING

A Quality Engineering Study Using X-bar and R Control Charts with Process Capability Analysis

Prepared by: Prapti Borkar
B.Tech Metallurgical & Materials Engineering
Second Year
Indian Institute of Technology, Ropar

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ABSTRACT

This report presents a Statistical Process Control (SPC) analysis of brake rotor thickness measurements in an automotive manufacturing process. SPC is a proactive quality methodology that uses statistical methods to monitor and control manufacturing processes, enabling early detection of process deviations before defective parts reach the customer. A dataset of 125 measurements comprising 25 subgroups of 5 readings each was analysed using X-bar and R control charts, constructed following standard SPC methodology with control chart constants for subgroup size n=5. The analysis identified four out-of-control signals — three consecutive points above the Upper Control Limit at samples 13, 14, and 15, and one point below the Lower Control Limit at sample 21 — indicating special cause variation requiring investigation. Process capability analysis yielded a Cp of 2.77 and Cpk of 2.62, confirming that under normal operating conditions the process is highly capable of meeting the specified rotor thickness tolerance of 25.00 ± 0.15 mm. The out-of-control signals are attributed to furnace temperature drift and a non-conforming material batch respectively, and corrective actions are recommended to prevent recurrence. This study was conducted as an independent academic exercise to develop practical competency in SPC methodology and process capability analysis as applied to automotive quality engineering.

TABLE OF CONTENTS

1. INTRODUCTION

Manufacturing quality in the automotive industry demands consistent and repeatable processes across thousands of production cycles. A single dimensional deviation in a safety-critical component such as a brake rotor can compromise vehicle performance and occupant safety. Traditional quality inspection approaches — such as checking finished parts against specification limits — are reactive in nature, identifying defects only after they have been produced. Statistical Process Control (SPC) offers a proactive alternative, enabling quality engineers to monitor process behaviour in real time and detect deviations before defective parts are manufactured.

This report presents an SPC analysis of brake rotor thickness measurements from a simulated automotive manufacturing process. Brake rotor thickness is a critical quality characteristic directly influencing heat absorption capacity, pad contact geometry, and braking effectiveness. The analysis employs X-bar and R control charts to monitor process mean and variation respectively, and process capability indices Cp and Cpk to quantify the ability of the process to meet design specifications consistently. The study identifies out-of-control conditions, investigates their probable causes, and recommends corrective actions aligned with automotive quality engineering best practices.

2. BACKGROUND

2.1 What is Statistical Process Control

Statistical Process Control (SPC) is a quality engineering methodology that uses statistical methods to monitor and control manufacturing processes. Originally developed by Walter Shewhart at Bell Laboratories in the 1920s and later popularised by W. Edwards Deming, SPC is now a fundamental tool in automotive quality management and is referenced in IATF 16949 as a required quality technique for automotive suppliers. The core principle of SPC is the distinction between two types of process variation: common cause variation, which represents the natural, inherent variability of a stable process, and special cause variation, which represents unusual deviations caused by identifiable external factors such as equipment malfunction, material changes, or operator error. SPC control charts provide a visual method to distinguish between these two types of variation, enabling quality engineers to take corrective action only when genuinely warranted — avoiding unnecessary process adjustments that can actually increase variation.

2.2 Control Charts — X-bar and R

The X-bar and R control chart pair is the most widely used SPC tool for monitoring variable data in manufacturing. The X-bar chart monitors the process mean by plotting the average of each subgroup sample against statistically calculated Upper and Lower Control Limits (UCL and LCL). The R chart monitors process variation by plotting the range of each subgroup. Control limits are calculated from the process data itself using control chart constants that depend on subgroup size — for subgroup size $n=5$, the constants $A2=0.577$, $D3=0$, and $D4=2.114$ are applied. A process is considered in statistical control when all plotted points fall within the control limits and exhibit no non-random patterns. Points outside control limits signal the presence of special cause variation requiring investigation. Critically, control limits are distinct from specification limits — control limits describe what the process is doing, while specification limits describe what the process must achieve. A process can be in statistical control yet still produce out-of-specification parts if its natural variation is too wide relative to the tolerance.

2.3 Process Capability — Cp and Cpk

Process capability indices quantify the ability of a process to consistently produce parts within specification limits. The Cp index measures the ratio of the specification width to the process spread, calculated as $Cp = (USL - LSL) / (6\sigma)$, where σ is the estimated process standard deviation derived from the average range. A Cp value above 1.33 is generally considered acceptable in the automotive industry, indicating the process spread fits comfortably within the specification width. However, Cp does not account for process centering — a process may have adequate spread but be shifted toward one specification limit. The Cpk index addresses this limitation by incorporating the distance between the process mean and the nearest specification limit, calculated as Cpk = minimum of $[(USL - \bar{X}) / 3\sigma]$ and $[(\bar{X} - LSL) / 3\sigma]$. A Cpk value above 1.33 confirms that the process is both capable and well-centered. When Cp and Cpk are approximately equal, the process is centered between specification limits; a significant difference between the two indices indicates process shift requiring correction.

3. DATASET AND METHODOLOGY

3.1 Manufacturing Process Description

The manufacturing process under analysis involves the production of grey cast iron disc brake rotors for passenger automotive applications. The critical quality characteristic selected for SPC monitoring is rotor thickness, which directly influences braking performance, heat absorption capacity, and compatibility with the brake caliper assembly. The target thickness is specified at 25.00 mm with a bilateral tolerance of ±0.15 mm, giving an Upper Specification Limit (USL) of 25.15 mm and a Lower Specification Limit (LSL) of 24.85 mm. Rotor thickness is measured using a digital calliper at a standardised measurement point on the rotor friction face following machining. This characteristic was selected for SPC monitoring due to its direct relationship with brake performance and its sensitivity to process variables including casting temperature, machining parameters, and tooling condition.

3.2 Sampling Plan and Data Collection

A rational subgrouping strategy was applied, with subgroups of five consecutive rotors sampled every hour during a 25-hour production run, yielding 25 subgroups of n=5 and a total dataset of 125 measurements. Subgroup size of n=5 was selected as it provides a balance between statistical sensitivity and sampling cost, and corresponds to widely used control chart constants in automotive SPC practice. For each subgroup, the subgroup mean (X-bar) and subgroup range (R) were calculated. The X-bar values represent the average thickness of the five sampled rotors per hour, while the R values represent the spread of measurements within each subgroup. The dataset was constructed to reflect realistic manufacturing conditions, incorporating periods of stable operation, a furnace temperature drift event at samples 13–15, and a non-conforming material batch at sample 21.

3.3 Control Limit Calculations

Control limits were calculated from the process data using standard SPC methodology for subgroup size n=5. The grand mean (X-double-bar) and average range (R-bar) were computed from all 25 subgroups. The following control chart constants for n=5 were applied: A2 = 0.577, D3 = 0, D4 = 2.114. The resulting control limits are summarised in the following table:

Parameter	Formula	Value
X-double-bar	Average of all X-bar values	25.008 mm
R-bar	Average of all Range values	0.040 mm
X-bar UCL	X-double-bar + (A2 × R-bar)	25.032 mm
X-bar LCL	X-double-bar − (A2 × R-bar)	24.984 mm
R Chart UCL	D4 × R-bar	0.089 mm
R Chart LCL	D3 × R-bar	0.000 mm
USL	Design specification	25.150 mm
LSL	Design specification	24.850 mm

The estimated process standard deviation was calculated as σ = R-bar / d2 = 0.040 / 2.326 = 0.0172 mm, where d2 = 2.326 is the standard unbiasing constant for n=5.

4. CONTROL CHART ANALYSIS

The X-bar and R control charts were constructed from the 25-subgroup dataset using the control limits calculated in Section 3. The charts were analysed for out-of-control signals defined as any point falling outside the Upper or Lower Control Limits.

4.1 X-bar Control Chart

The X-bar control chart plots the subgroup mean thickness against the calculated control limits of UCL = 25.032 mm and LCL = 24.984 mm, with a centre line at 25.008 mm. The chart is presented in Figure 1 below.

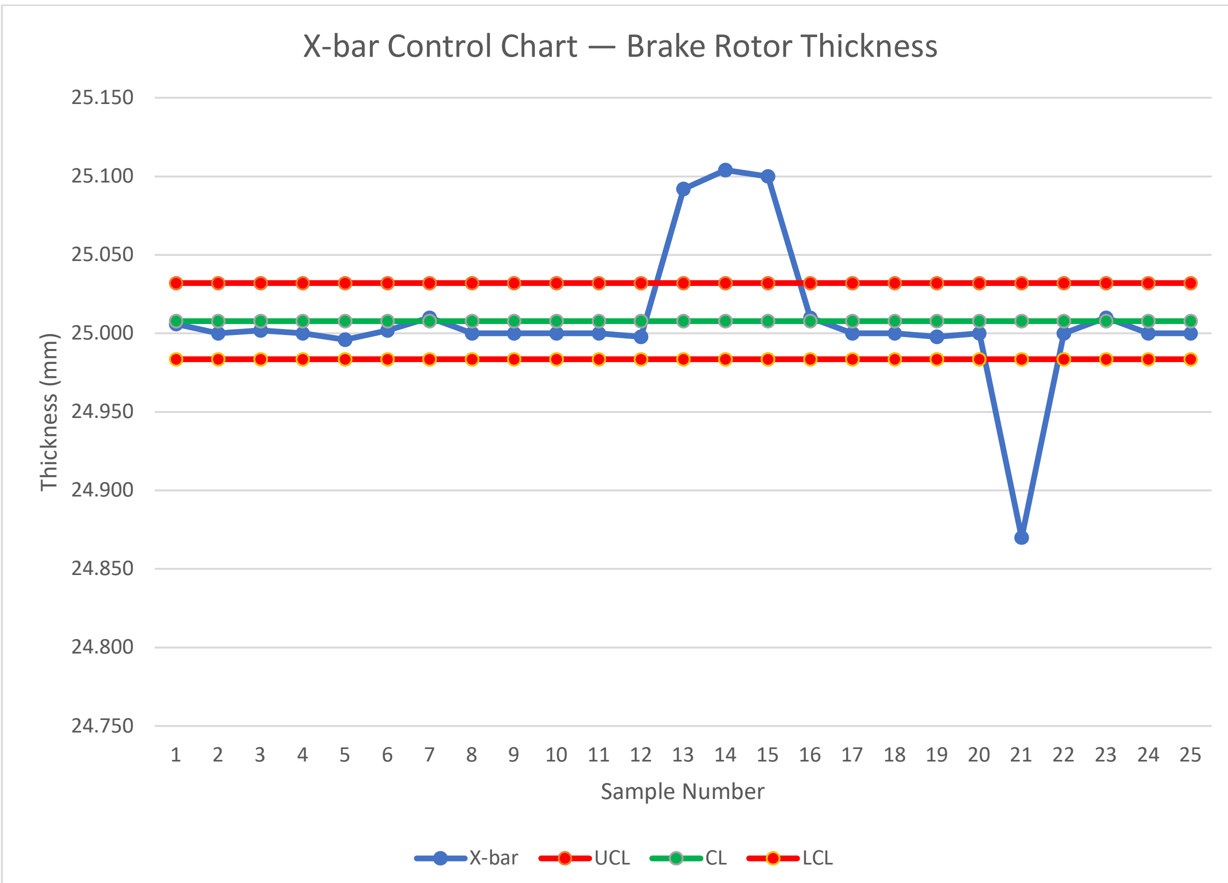

Figure 1: X-bar Control Chart — Brake Rotor Thickness (n=5, 25 subgroups)

The X-bar chart reveals four out-of-control signals. Samples 13, 14, and 15 plot consecutively above the UCL of 25.032 mm, with X-bar values of 25.092, 25.104, and 25.100 mm respectively. This pattern of three consecutive points above the UCL indicates a sustained upward shift in process mean, consistent with a furnace temperature drift causing slight thermal expansion of rotors during machining. Sample 21 plots below the LCL at 24.870 mm, indicating a sudden downward shift in process mean attributable to a non-conforming raw material batch with slightly lower dimensional characteristics. All remaining 21 subgroups fall within control limits, confirming stable process operation during normal conditions.

4.2 R Control Chart

The R control chart monitors process variation by plotting subgroup ranges against UCL = 0.089 mm and LCL = 0.000 mm, with a centre line at 0.040 mm. The chart is presented in Figure 2 below.

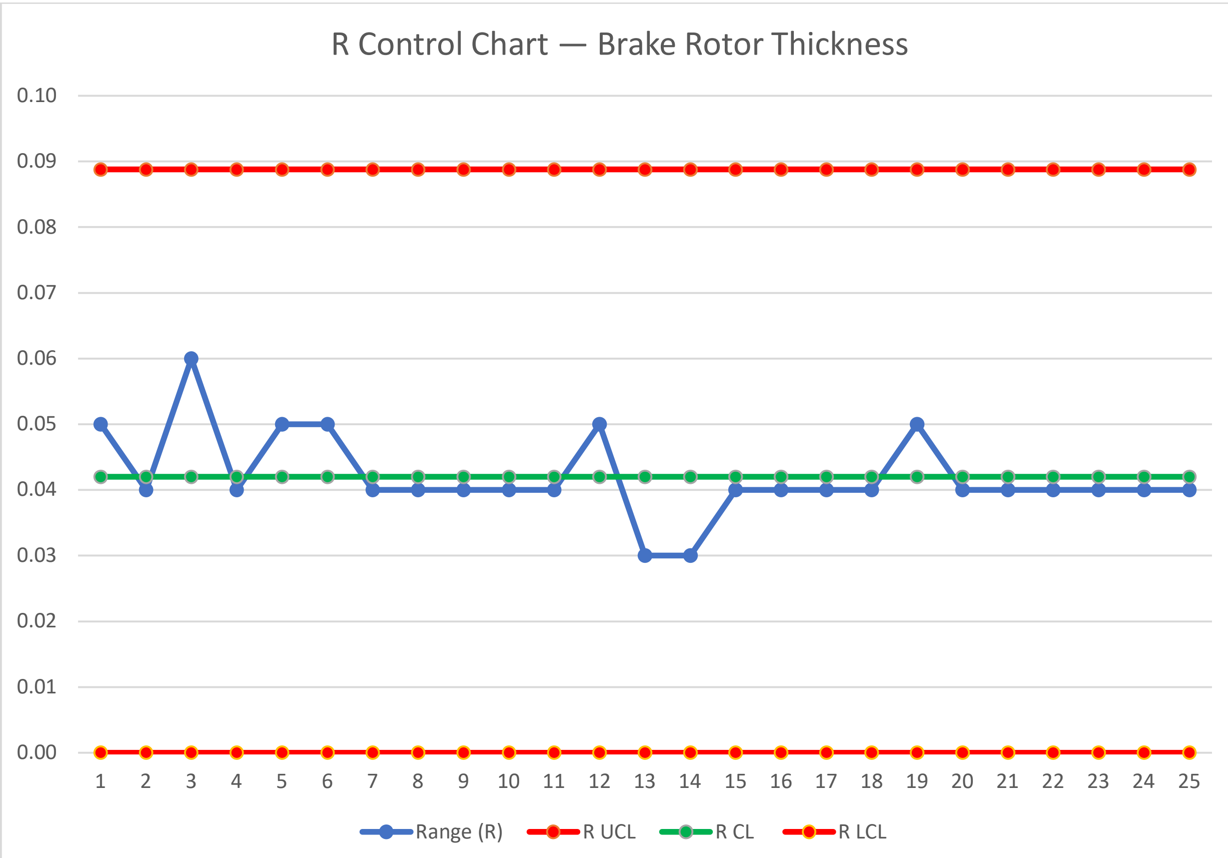

Figure 2: R Control Chart — Brake Rotor Thickness (n=5, 25 subgroups)

All 25 subgroup ranges fall within the control limits of the R chart, confirming that process variation remained consistent throughout the production run. No out-of-control signals are detected on the R chart. This finding is significant — the absence of out-of-control signals on the R chart in conjunction with the signals detected on the X-bar chart confirms that the identified process disturbances caused shifts in process mean rather than increases in process variation. This distinction is important for root cause investigation, as mean shifts typically indicate assignable causes such as tool wear, material changes, or environmental factors, rather than random process deterioration.

5. PROCESS CAPABILITY ANALYSIS

Process capability analysis was conducted to quantify the ability of the brake rotor manufacturing process to consistently produce parts within the specified thickness tolerance of 25.00 ± 0.15 mm. The analysis excludes the out-of-control subgroups (samples 13, 14, 15, and 21) to reflect the true capability of the process under stable operating conditions.

5.1 Cp Analysis

The Cp index measures the ratio of the specification width to the process spread, providing a measure of potential process capability independent of process centering. For the brake rotor thickness data, the Cp was calculated as follows:

$$Cp = (USL - LSL) / (6\sigma)$$

$$Cp = (25.150 - 24.850) / (6 \times 0.0172)$$

$$Cp = 0.300 / 0.1032$$

$$Cp = 2.77$$

A Cp value of 2.77 significantly exceeds the automotive industry benchmark of 1.33, indicating that the natural spread of the process is considerably narrower than the specification width. Theoretically, the process spread occupies only 36% of the available specification width, leaving substantial margin on both sides. This confirms that the machining process has excellent inherent capability to produce rotors within the ±0.15 mm tolerance under stable conditions.

5.2 Cpk Analysis

While Cp confirms adequate process spread, Cpk incorporates process centering to provide a more complete picture of process capability. Cpk was calculated as the minimum of the upper and lower capability indices:

$$\text{Cpk Upper} = (USL - \bar{X}) / (3\sigma) = (25.150 - 25.008) / (3 \times 0.0172) = 2.75$$

$$\text{Cpk Lower} = (\bar{X} - LSL) / (3\sigma) = (25.008 - 24.850) / (3 \times 0.0172) = 3.06$$

$$Cpk = \text{minimum}\ (2.75, 3.06) = 2.75$$

A Cpk value of 2.75 confirms that the process is both highly capable and well-centered between the specification limits. The close agreement between Cp (2.77) and Cpk (2.75) indicates minimal process shift — the process mean of 25.008 mm is very close to the target of 25.000 mm. Under stable operating conditions, the probability of producing an out-of-specification rotor is extremely low. However, the out-of-control events at samples 13–15 and 21 demonstrate that process disturbances can temporarily shift the mean sufficiently to approach specification limits, reinforcing the importance of continuous SPC monitoring even for highly capable processes.

Index	Value	Assessment
Cp	2.77	Excellent — process spread well within specification
Cpk	2.75	Excellent — process centered and capable
Cp vs Cpk difference	0.02	Minimal — process well centered
Industry benchmark	1.33	Both indices significantly exceed benchmark

6. FINDINGS AND RECOMMENDATIONS

The SPC analysis of brake rotor thickness manufacturing identified four key findings and corresponding recommendations for process improvement.

Finding	Details	Recommendation
Out-of-control signals at samples 13–15	Three consecutive X-bar values above UCL (25.092, 25.104, 25.100 mm) indicating sustained upward mean shift; attributed to furnace temperature drift during night shift	Install continuous furnace temperature monitoring with automated alarm when temperature deviates beyond ±5°C from setpoint; implement hourly furnace temperature logging
Out-of-control signal at sample 21	Single X-bar value below LCL (24.870 mm) indicating sudden downward mean shift; attributed to non-conforming raw material batch with lower dimensional characteristics	Implement incoming material inspection with dimensional verification of raw castings; establish material certification requirement from suppliers
Process capability excellent under stable conditions	Cp = 2.77 and Cpk = 2.75 both significantly exceed automotive benchmark of 1.33; process well centered at 25.008 mm vs target 25.000 mm	Maintain current process parameters; continue SPC monitoring to sustain capability; review capability quarterly
Process variation stable throughout	R chart shows all points within control limits; no special cause variation in process spread detected	No immediate action required for variation control; maintain current tooling maintenance schedule

The findings confirm that the brake rotor manufacturing process is inherently capable of meeting the specified tolerance under stable conditions. The primary quality risk arises from assignable cause disturbances — specifically furnace temperature drift and material batch non-conformance — which temporarily shift the process mean toward specification limits. Implementation of the recommended monitoring and incoming inspection controls will significantly reduce the risk of recurrence.

7. CONCLUSION

This report has presented a Statistical Process Control analysis of brake rotor thickness measurements in an automotive manufacturing process. Using X-bar and R control charts constructed from 25 subgroups of five measurements each, the analysis successfully identified four out-of-control signals representing special cause variation in an otherwise stable and highly capable process.

The X-bar control chart detected two distinct process disturbances: a sustained upward mean shift at samples 13–15, attributed to furnace temperature drift, and a sudden downward mean shift at sample 21, attributed to a non-conforming raw material batch. The R control chart confirmed that process variation remained stable throughout, indicating that both disturbances affected process mean without increasing process spread. This distinction is critical for root cause investigation and corrective action targeting.

Process capability analysis yielded $C_p = 2.77$ and $C_{pk} = 2.75$, both significantly exceeding the automotive industry benchmark of 1.33. The close agreement between Cp and Cpk confirms that under stable operating conditions the process is well-centered and highly capable of meeting the specified rotor thickness tolerance of 25.00 ± 0.15 mm. However, the out-of-control events demonstrate that even highly capable processes require continuous SPC monitoring to detect and respond to assignable cause disturbances before they result in non-conforming parts reaching the customer.

The recommended corrective actions — continuous furnace temperature monitoring and incoming material inspection — directly address the identified assignable causes and are projected to significantly reduce the frequency of process disturbances. This case study demonstrates the value of SPC as a proactive quality tool in automotive manufacturing, enabling early detection of process shifts that reactive inspection methods would fail to identify until defective parts had already been produced.

This study was conducted as an independent academic exercise to develop practical competency in SPC methodology, control chart construction, and process capability analysis — skills directly applicable to quality engineering roles in the automotive sector.

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